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Class: Signals and Systems

Lab #: 07

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Lab 7: Noise Reduction

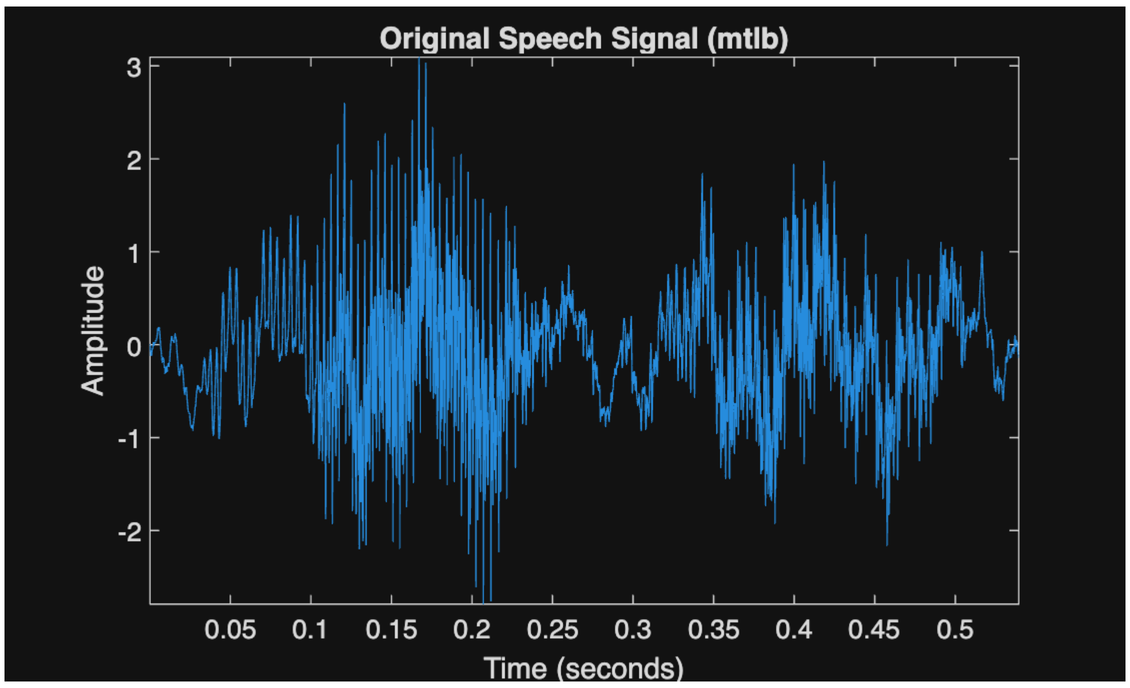
```
%  ELLIPTIC LOW-PASS FILTER DESIGN AND SIGNAL ANALYSIS
%  This script:
%   1. Loads and plots the original MATLAB speech signal (mtlb)
%   2. Loads and plots a noisy speech signal
%   3. Compares their spectra
%   4. Designs and applies an elliptic low-pass filter
%   5. Displays results and responses
```

1. SETUP PARAMETERS

```
Fs = 7418;           % Sampling frequency (Hz)
```

2. LOAD AND PLOT ORIGINAL SIGNAL

```
load mtlb;            % Load built-in MATLAB speech signal
L = length(mtlb);     % Length of the signal

figure;
plot((1:L)/Fs, mtlb);
axis tight;
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Original Speech Signal (mtlb)');
```

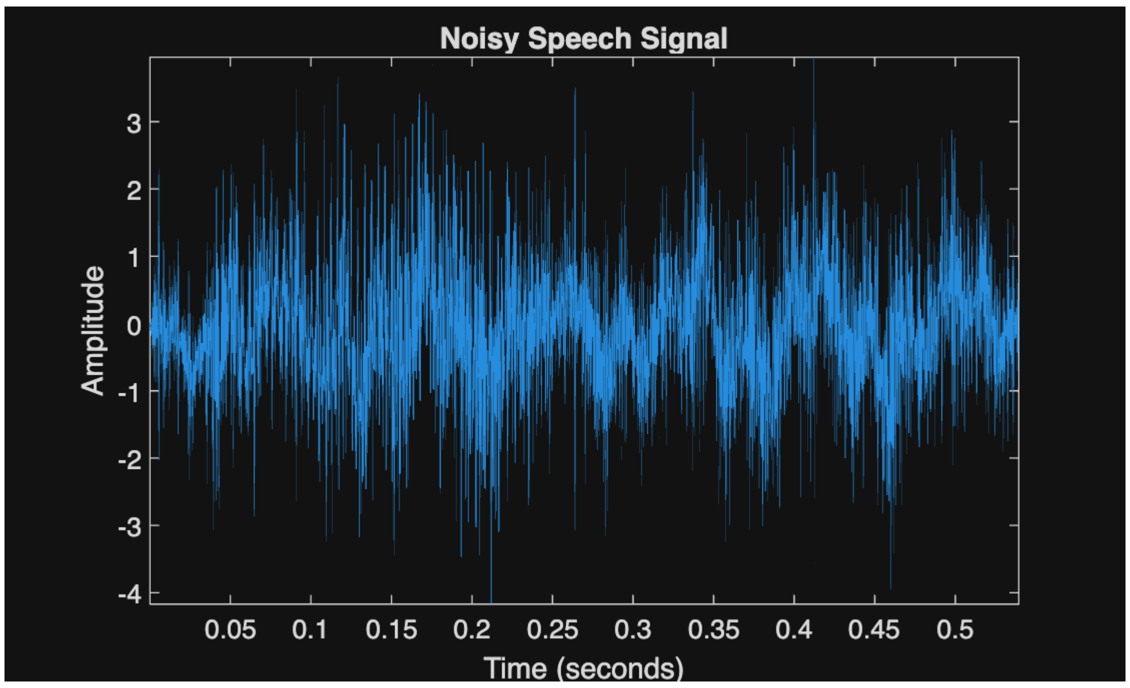



```
% Play the original audio
soundsc(mtlb, Fs);
```

3. LOAD AND PLOT NOISY SIGNAL

```
x = load('NoisySpeech.txt');   % Load noisy speech from file

figure;
plot((1:length(x))/Fs, x);
axis tight;
xlabel('Time (seconds)');
ylabel('Amplitude');
```

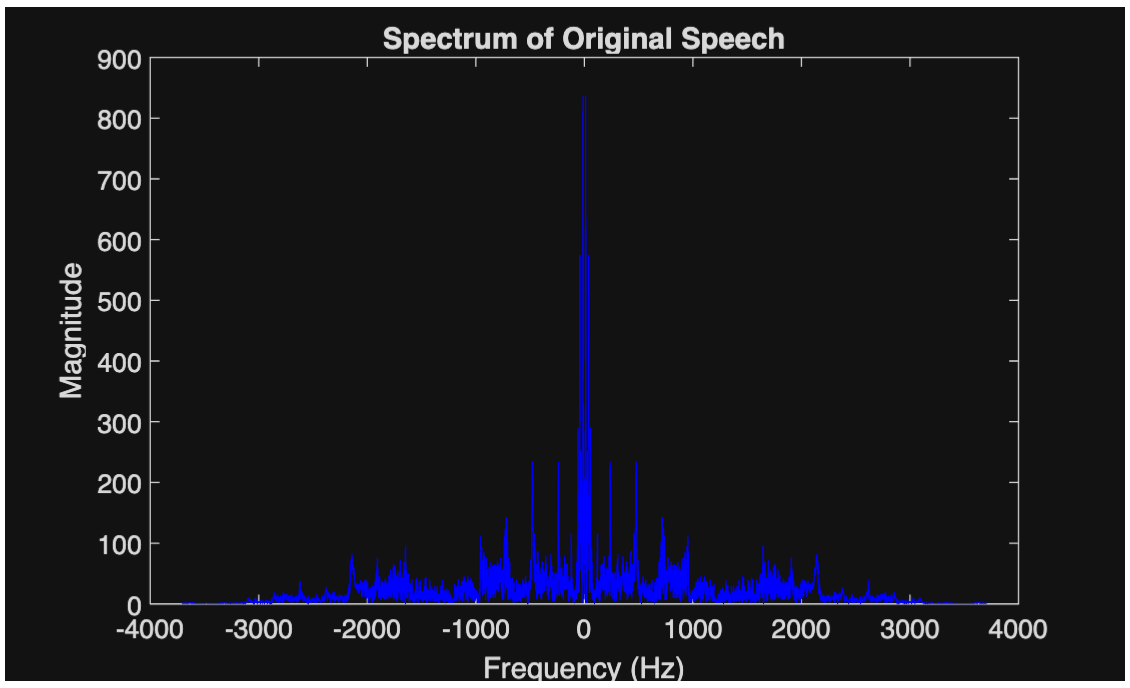
```
title('Noisy Speech Signal');
```




```
% Play the noisy audio
soundsc(x, Fs);
```

4. PLOT FREQUENCY SPECTRUM

```
figure;
% Compare the spectra of original and noisy signals
[M1, f1] = dtft(mtlb, 1/Fs);
[M2, f2] = dtft(x, 1/Fs);

figure;
plot(f1, M1, 'b');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Original Speech');
```




```
disp('Most of the energy in the mtlb speech signal is concentrated below about 2.5 kHz.');
```

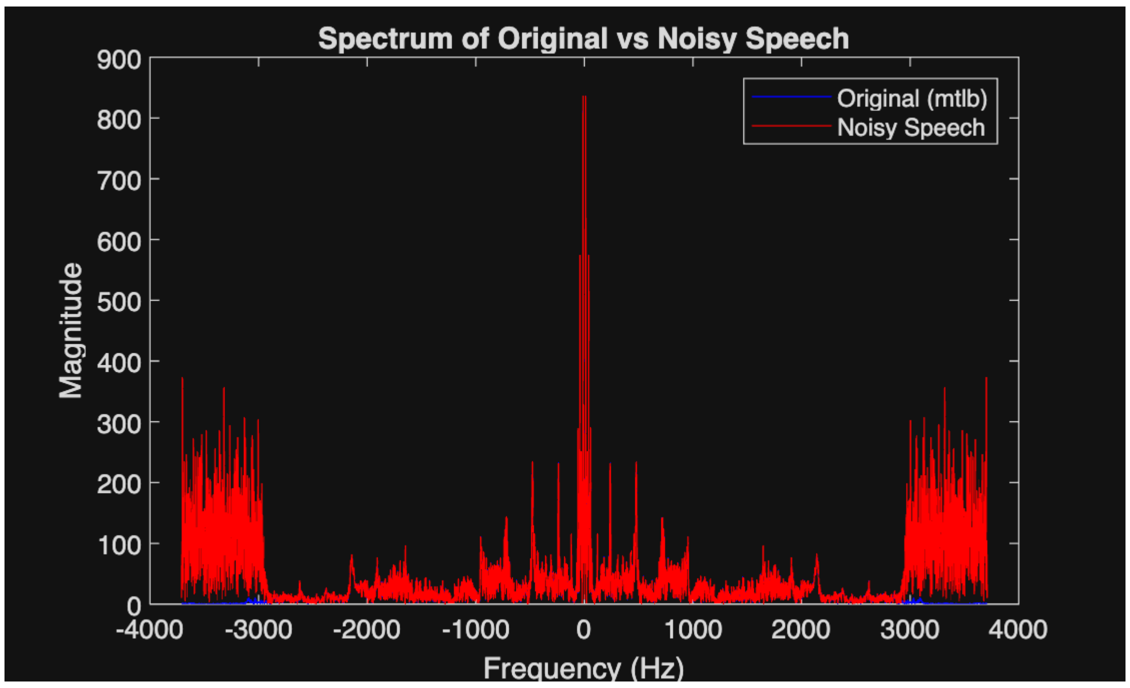
```
Most of the energy in the mtlb speech signal is concentrated below about 2.5 kHz.
```

```
disp('The strongest components occur mainly below 2 kHz, while higher frequencies above about 3 kHz contain
only weak speech energy.');
```

```
The strongest components occur mainly below 2 kHz, while higher frequencies above about 3 kHz contain only weak speech energy.
```

```
% Compare the spectra of original and noisy signals

[M1, f] = dtft(mtlb, 1/Fs);
figure;
plot(f, M1, 'b');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Original vs Noisy Speech');
hold on;
```

```
plot(f, M2, 'r');
legend('Original (mtlb)', 'Noisy Speech');
hold off;
```




```
disp('When comparing the spectra of the original and noisy signals, the extra energy at high frequencies in the ');
```

When comparing the spectra of the original and noisy signals, the extra energy at high frequencies in the

```
disp('noisy signal represents the additive noise.');
```

noisy signal represents the additive noise.

```
disp('Based on the spectrum, the noise is mainly concentrated above 2.7 kHz.');
```

Based on the spectrum, the noise is mainly concentrated above 2.7 kHz.

```
disp('The low-frequency region where both spectra coincide corresponds to the actual speech content, while the high-frequency region, where');
```

The low-frequency region where both spectra coincide corresponds to the actual speech content, while the high-frequency region, where

```
disp('the noisy signals magnitude rises above that of the clean signal, is dominated by noise.');
```

the noisy signals magnitude rises above that of the clean signal, is dominated by noise.

5. DESIGN ELLIPTIC LOW-PASS FILTER

```
disp('A lowpass filter allows the low-frequency components (which contain the essential speech information) to pass through while');
```

A lowpass filter allows the low-frequency components (which contain the essential speech information) to pass through while

```
disp('attenuating the high-frequency components where the noise dominates.');
```

attenuating the high-frequency components where the noise dominates.

```
disp('By suppressing these higher frequencies, the filter reduces the noise power, making the speech clearer without greatly affecting');
```

By suppressing these higher frequencies, the filter reduces the noise power, making the speech clearer without greatly affecting

```
disp('its intelligibility.');
```

its intelligibility.

```
disp('The cut-off frequency should be chosen just above the highest significant speech frequency, where the speech energy starts to');
```

The cut-off frequency should be chosen just above the highest significant speech frequency, where the speech energy starts to

```
disp('fall off and the noise becomes dominant.');
```

fall off and the noise becomes dominant.

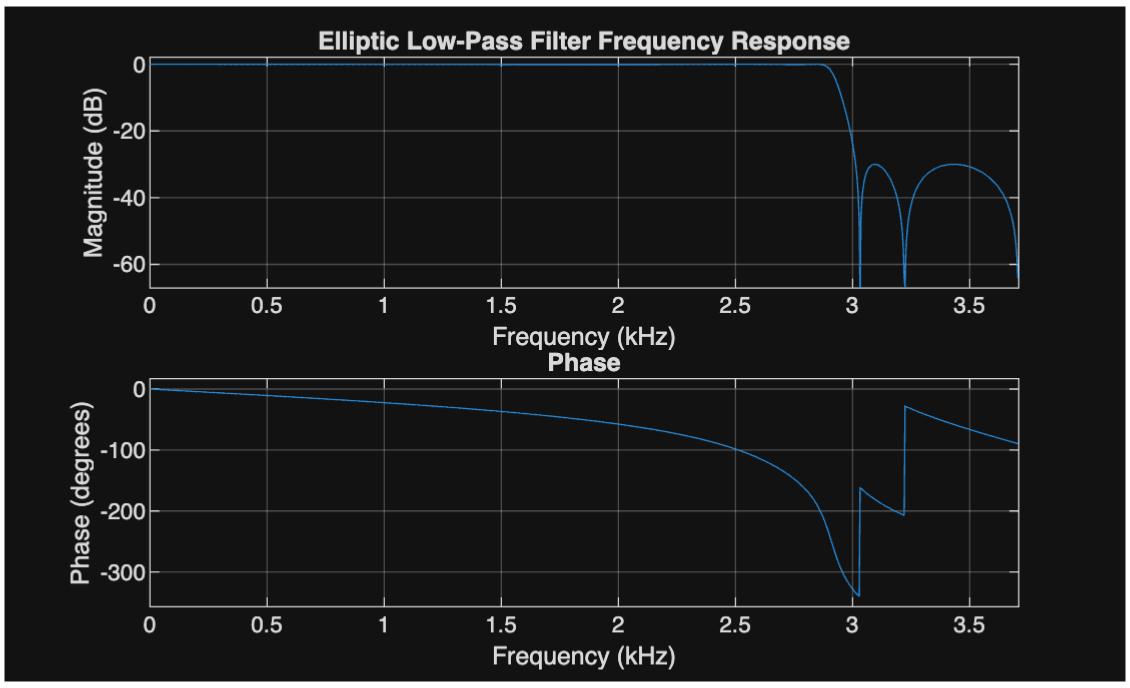
```
disp('From observation of the spectrum, a cut-off around 2.8 – 3.0 kHz provides a good balance – preserving the speech band while removing');
```

From observation of the spectrum, a cut-off around 2.8 – 3.0 kHz provides a good balance – preserving the speech band while removing

```
disp('most of the high-frequency noise.');
```

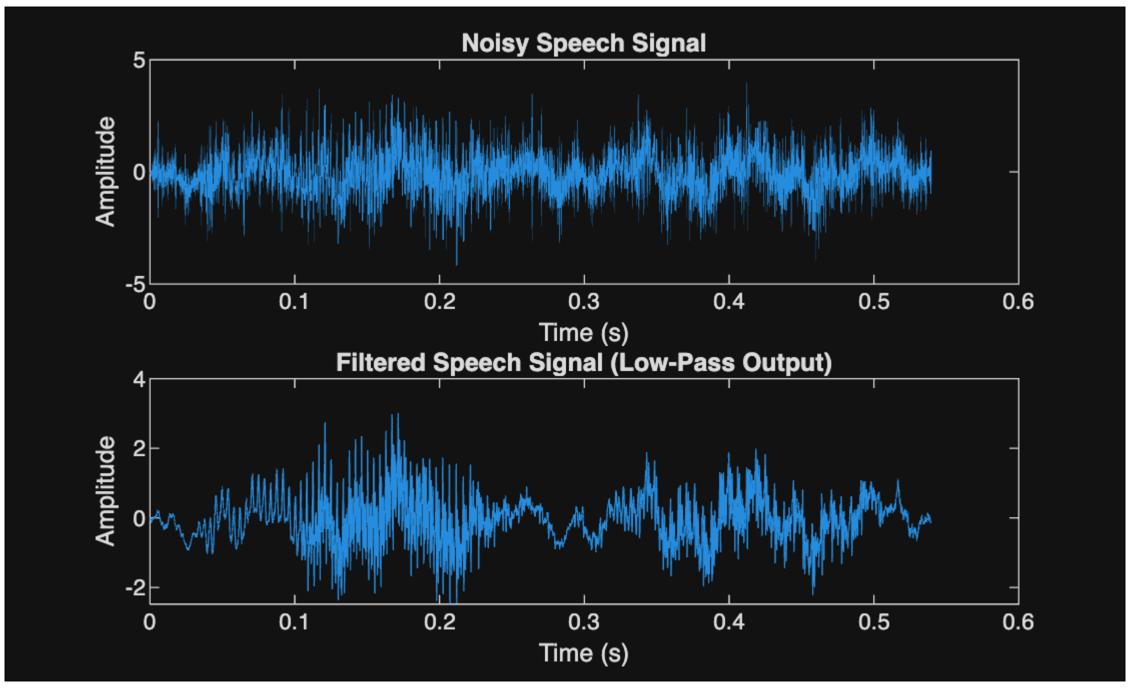
most of the high-frequency noise.

```
% Filter specifications
Fc = 2867.5;           % Cutoff frequency (Hz)
Rp = 0.1;              % Passband ripple (dB)
Rs = 30;              % Stopband attenuation (dB)
n  = 5;              % Filter order

% Normalized cutoff frequency (relative to Nyquist)
Wn = Fc / (Fs/2);

% Design elliptic low-pass filter
[b, a] = ellip(n, Rp, Rs, Wn, 'low');

% Plot the frequency response of the filter
figure;
freqz(b, a, 1024, Fs);
title('Elliptic Low-Pass Filter Frequency Response');
```



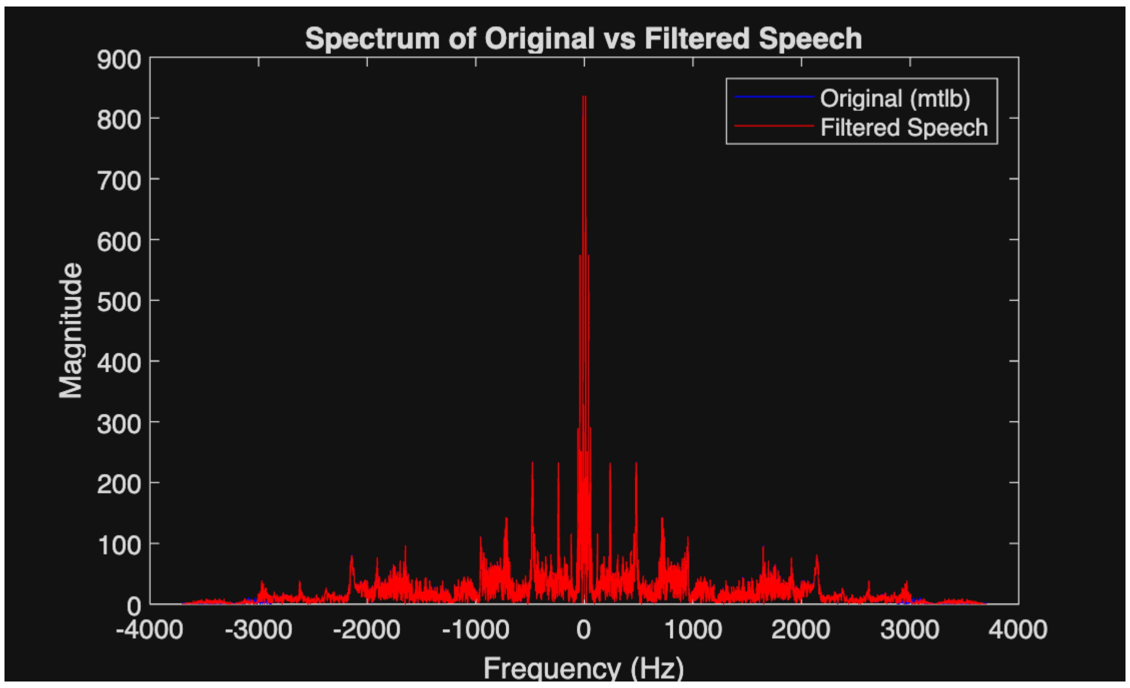
6. APPLY FILTER TO NOISY SIGNAL

```
y = filter(b, a, x);   % Filter the noisy signal

% Play the filtered (denoised) speech
soundsc(y, Fs);

% Compare noisy and filtered signals in time domain
figure;
subplot(2,1,1);
plot((1:length(x))/Fs, x);
xlabel('Time (s)');
ylabel('Amplitude');
title('Noisy Speech Signal');

subplot(2,1,2);
plot((1:length(y))/Fs, y);
xlabel('Time (s)');
ylabel('Amplitude');
title('Filtered Speech Signal (Low-Pass Output)');
```




```
% Compare the spectra of original and filtered signals
```

```
[M1, f] = dtft(mtlb, 1/Fs);
figure;
plot(f, M1, 'b');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Spectrum of Original vs Filtered Speech');
hold on;

[M4, f] = dtft(y, 1/Fs);
plot(f, M4, 'r');
legend('Original (mtlb)', 'Filtered Speech');
hold off;
```




```
disp('Using an elliptic lowpass filter with a cut-off near 2.9 kHz, most of the high-frequency noise is
removed, producing a noticeably');
```

Using an elliptic lowpass filter with a cut-off near 2.9 kHz, most of the high-frequency noise is removed, producing a noticeably

```
disp('clearer signal.');
```

clearer signal.

```
disp('However, because the noise overlaps partially with the speech band, some residual noise remains, and
excessive filtering can');
```

However, because the noise overlaps partially with the speech band, some residual noise remains, and excessive filtering can

```
disp('make the speech sound dull or muffled.');
```

make the speech sound dull or muffled.

```

disp('The stop-band ripple (or stop-band attenuation) has a greater effect on noise reduction quality.');
```

The stop-band ripple (or stop-band attenuation) has a greater effect on noise reduction quality.

```
disp('A larger stop-band attenuation (smaller ripple) suppresses more of the unwanted high-frequency noise.');
```

A larger stop-band attenuation (smaller ripple) suppresses more of the unwanted high-frequency noise.

```
disp('The passband ripple mainly affects the naturalness and fidelity of the speech but has a smaller impact
on overall noise level.');
```

The passband ripple mainly affects the naturalness and fidelity of the speech but has a smaller impact on overall noise level.

```
disp('Therefore, in this application it is more important that the stopband be close to 0 (strong attenuation)
than that the passband be');
```

Therefore, in this application it is more important that the stopband be close to 0 (strong attenuation) than that the passband be

```
disp('perfectly flat.');
```

perfectly flat.

```

disp('The elliptic lowpass filter designed with parameters Fc ≈ 2867.5 Hz, Rp = 0.1 dB, Rs = 30 dB, order = 5
successfully reduces');
```

The elliptic lowpass filter designed with parameters Fc ≈ 2867.5 Hz, Rp = 0.1 dB, Rs = 30 dB, order = 5 successfully reduces

```
disp('high-frequency noise while maintaining speech intelligibility.');
```

high-frequency noise while maintaining speech intelligibility.

```
disp('Increasing stopband attenuation or slightly lowering the cutoff frequency further improves noise
suppression at the cost of some loss');
```

```
Increasing stopband attenuation or slightly lowering the cutoff frequency further improves noise suppression at the cost of some loss
```

```
disp('in brightness.');
```

```
in brightness.
```